

A NOTE ON VARIABLE SELECTION IN LINEAR MODELS

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Summary: In this note a new discrepancy, the partial Gauss-discrepancy, for variable selection in linear models is proposed. The discrepancy is applicable when the estimation of a subset of the parameters in a linear model is of interest, and reduces to the usual Gauss-discrepancy when estimation of the whole parameter vector is of interest. The corresponding variable selection criterion is derived, which generalizes criteria of Mallows (1973) and Linhart and Zucchini (1986). This criterion can be applied in linear regression, analysis of variance, and analysis of covariance. An interesting relationship between the criterion for variable selection resulting from the partial Gauss-discrepancy, a generalization of Cook's (1977) distance for variables, and a generalized variance inflation factor emerges.

1. INTRODUCTION

We consider the general linear model

$$(1.1) \quad y = [X : Z] \begin{bmatrix} \beta^* \\ \gamma^* \end{bmatrix} + e$$

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where $y_{n \times 1}$ is a vector of observations, $X_{n \times q}$ and $Z_{n \times r}$ are known design matrices with $q + r = k$, $\beta_{q \times 1}^*$ and $\gamma_{r \times 1}^*$ are parameter vectors, and $e_{n \times 1}$ is a vector of residuals with $E(e) = 0$ and $\text{cov}(e) = \sigma^2 I_n$. For simplicity we assume initially that the matrix $[X : Z]$ is of full column rank. We assume that (1.1) is the *operating model* (Linhart and Zucchini, 1986, pp. 1-2, whose terminology we use throughout) for the data y . We suppose that the estimation of the parameters β^* is of primary interest, and we wish to decide whether the estimate of β^* is improved if some of the variables X or Z are omitted from the operating model. The problem is to select that subset of the variables $[X : Z]$ which leads to the most precise estimate for β^* , where precision is measured by the *discrepancy* to be defined below.

Situations where interest focuses on a parameter subset of a statistical model arise frequently in practice. In analysis of covariance, estimation of the treatment or design effects may be of interest, while the parameters associated with the covariates are only fitted to avoid bias in the estimates of the treatment parameters (Linhart and Zucchini, 1982 and 1986, ch. 9). For example, in a clinical trial one might be interested to estimate the difference in response between a treatment and a control group, while adjusting for patients' age, sex, and other covariates. Thus in analysis of covariance, the matrix X in (1.1) represents the treatment or design effects, while Z represents the covariates. Similarly, a parameter subset may be the focus in a linear regression model. In analysis of variance, the variables Z may represent block effects or stratifying factors which are not of intrinsic interest to the analysis, while the matrix X represents the treatment or design effects of interest.

In this note we propose a discrepancy, the *partial Gauss-discrepancy*, for model selection in the situation that a parameter subset of the operating model is the focus of interest. We derive a *criterion*, that is, an unbiased estimate of the expected discrepancy. The model which is ultimately selected is that which minimizes the criterion. When the whole parameter vector is of interest, the discrepancy reduces to the usual Gauss-discrepancy, and the criterion is equivalent to Mallows' C_p criterion for variable selection in linear regression, and to the criterion of Linhart and Zucchini (1986, ch.

8) for variable selection in analysis of variance. Thus the partial Gauss-discrepancy generalizes the Gauss-discrepancy, and the resulting criterion generalizes the criteria of Mallows (1973) and Linhart and Zucchini (1986, ch. 7-9) to the situation that the estimation of a subset of the model parameters is of interest. The proposed methodology can be applied to analysis of variance, analysis of covariance, and linear regression.

An interesting relationship between the variable selection criterion resulting from the partial Gauss-discrepancy, a generalization of Cook's (1977) distance for variables, and a generalized variance inflation factor emerges.

2. THE PARTIAL GAUSS-DISCREPANCY

Fitting all variables of the operating model (1.1) does not necessarily lead to the most precise estimate for β^* . In general the estimate for β^* is improved if some of the variables $[X : Z]$ are omitted from the operating model. The problem is to select a subset of $0 \leq q_1 \leq q$ variables X_1 from the variables X , and a subset of $0 \leq r_1 \leq r$ variables Z_1 from variables Z , such that the precision of the estimates of the parameters β^* is maximized. Partitioning the model (1.1) conformably as

$$y = [X_1 : X_2 : Z_1 : Z_2] \begin{bmatrix} \beta_1^* \\ \beta_2^* \\ \gamma_1^* \\ \gamma_2^* \end{bmatrix} + e$$

an *approximating model* is in general of the form

$$(2.1) \quad y = [X_1 : Z_1] \begin{bmatrix} \beta_1 \\ \gamma_1 \end{bmatrix} + e = S_1 \tau_1 + e$$

where $S_1 = [X_1 : Z_1]$ is a matrix of $p = q_1 + r_1$ variables, and $\tau_1 = [\beta_1', \gamma_1']'$. We denote the $q_2 + r_2$ variables and parameters which are dropped from the operating model by $S_2 = [X_2 : Z_2]$ and $\tau_2 = [\beta_2', \gamma_2']'$. Note that in (1.1) and (2.1) we distinguish between the parameters β^* and γ^* under the operating model, and the parameters β and γ under an approximating model (see Linhart and Zucchini, 1986, p. 158). Where this distinction is not necessary, the asterisks are dropped.

As a suitable discrepancy for the situation that the estimation of the parameter subset β^* is of interest we propose

$$(2.2) \quad \Delta(\beta) = (\beta - \beta^*)' M (\beta - \beta^*)$$

where M is a nonnegative definite matrix. Some possible choices of M are

$$(2.3) \quad M = I_q$$

$$(2.4) \quad M = X'X$$

$$(2.5) \quad M = X'X - X'Z(Z'Z)^{-1}Z'X.$$

The matrix (2.3) was suggested by Linhart and Zucchini (1982 and 1986, ch. 9). However, for $M = I_q$ the discrepancy is not invariant under linear reparametrizations of $X\beta^*$ of the form $X\beta^* = XUT\beta^* = \tilde{X}\tilde{\beta}^*$. Choosing M as in (2.4) or (2.5), this invariance property is satisfied. When M is chosen as in (2.5), then (2.2) is invariant under reparametrizations of the above form, and under reparametrizations of the form $X\beta^* + Z\gamma^* = (X - ZT)\beta^* + Z(\gamma^* + T\beta^*) = \tilde{X}\tilde{\beta}^* + Z\tilde{\gamma}^*$. Note that the least-squares estimate of β^* under the operating model is invariant under reparametrizations of that form. If one wants a discrepancy which is invariant under reparametrizations which leave the least-squares estimate of β^* invariant, then M must be chosen as in (2.5). Note also that M as in (2.5) is proportional to the inverse of the covariance matrix of the least-squares estimate $\hat{\beta}$ for β^* under the operating model. When M is chosen as in (2.5), we call (2.2) the *partial Gauss-discrepancy*, while for M as in (2.4) we call (2.2) the *marginal Gauss-discrepancy*.

When $q = p$ and thus $r = 0$, that is, the estimation of the whole parameter vector is of interest, both the marginal and the partial Gauss-discrepancy reduce to

$$\Delta(\beta) = (\beta - \beta^*)' X'X (\beta - \beta^*).$$

This is the usual Gauss-discrepancy. Thus both the marginal and the partial Gauss-discrepancy are generalizations of the usual Gauss-discrepancy. For $M = I_p$ discrepancy (2.2) reduces to the Gauss-discrepancy only when the variables X are orthonormal. While we consider (2.5) to be the canonical choice for M , we develop in the next section the criterion for variable selection for general M .

3. THE CRITERION

A criterion is an estimate of the expected overall discrepancy of an approximating model. Using discrepancy (2.2), the expected overall discrepancy is

$$(3.1) \quad E[(\tilde{\beta} - \beta^*)' M (\tilde{\beta} - \beta^*)]$$

where $\tilde{\beta} = [\tilde{\beta}_1', \tilde{\beta}_2']'$, and $\tilde{\beta}_1$ is the least-squares estimate for β_1 under the approximating model, and $\tilde{\beta}_2 = 0$. Thus (3.1) is the mean squared error of $\tilde{\beta}$, in the inner product norm with inner product matrix M . If one selects the model which minimizes an estimate of (3.1), then the model which leads to the minimal estimated mean-squared error of $\tilde{\beta}$ is selected. In that sense, it will yield the most precise estimate of β^* . As a general result one has a decomposition of the overall expected discrepancy (3.1) as

$$(3.2) \quad E[(\tilde{\beta} - \beta^*)' M (\tilde{\beta} - \beta^*)] = \text{trace}[\text{cov}(\tilde{\beta})M] + b' M b$$

where $b = E(\tilde{\beta} - \beta^*)$ is the bias of $\tilde{\beta}$. One has

$$\text{cov}(\tilde{\beta}_1) = \sigma^2 \cdot (X_1' X_1 - X_1' Z_1 (Z_1' Z_1)^{-1} Z_1' X_1)^{-1} = \sigma^2 \cdot (\tilde{X}_1' \tilde{X}_1)^{-1}$$

where $\tilde{X}_1 = (I_n - Z_1 (Z_1' Z_1)^{-1} Z_1') X_1$. Of course, $\text{cov}(\tilde{\beta}_1, \tilde{\beta}_2) = 0$, and $\text{cov}(\tilde{\beta}_2) = 0$. An unbiased estimate of the first term on the right hand side of (3.2) is thus given by

$$(3.3) \quad \text{trace}[\text{cov}(\tilde{\beta})M] = \hat{\sigma}^2 \cdot \text{trace}[(\tilde{X}_1' \tilde{X}_1)^{-1} M_{11}].$$

Here $\hat{\sigma}^2 = RSS_k / (n - k)$ is the residual mean-square under the operating model, and M_{11} is the leading principal $q_1 \times q_1$ submatrix of M . An unbiased estimate $\tilde{b}$ for the bias b is

$$(3.4) \quad \tilde{b} = \begin{bmatrix} \tilde{b}_1 \\ \tilde{b}_2 \end{bmatrix} = \begin{bmatrix} (\tilde{X}_1' \tilde{X}_1)^{-1} \tilde{X}_1' [X_2 : Z_2] \\ -I_{q_2} : 0 \end{bmatrix} \begin{bmatrix} \hat{\beta}_2 \\ \hat{\gamma}_2 \end{bmatrix} = B \hat{\tau}_2$$

where $\hat{\beta}_2$ and $\hat{\gamma}_2$ are the least-squares estimates of β_2 and γ_2 under the operating model. Since

$$E[\tilde{b}' M \tilde{b}] = b' M b + \text{trace}[\text{cov}(\tilde{b}) M],$$

an unbiased estimate $b' \widehat{M} b$ of the second summand on the right hand side of (3.2) is given by

$$(3.5) \quad \hat{\tau}_2' B' M B \hat{\tau}_2 - \hat{\sigma}^2 \cdot \text{trace}[B(S_2' S_2 - S_2' S_1 (S_1' X_1)^{-1} S_1' S_2)^{-1} B' M].$$

Using (3.3) and (3.5) in (3.2), the criterion for approximating model (2.1) is

$$(3.6) \quad \hat{\tau}_2' B' M B \hat{\tau}_2 + \hat{\sigma}^2 \cdot \text{trace} [(X_1' X_1 - X_1' Z_1 (Z_1' Z_1)^{-1} Z_1' X_1)^{-1} M_{11}] \\ - \hat{\sigma}^2 \cdot \text{trace} [(S_2' S_2 - S_2' S_1 (S_1' S_1)^{-1} S_1' S_2)^{-1} B' M B].$$

The approximating model which minimizes criterion (3.6) is selected for the estimation of the parameters β^* .

At this stage it may be noted that all quantities in (3.6) are well-defined even if X is not of full column rank. One needs only that the column spaces of X and Z , and the column spaces of S_1 and S_2 are disconnected. The former condition is satisfied if and only if $X\beta$ is estimable under the operating model, while for example in overparametrized factorial models the latter condition means that only hierarchical approximating models, obeying the principle of marginality, are fitted. When X or Z are not of full column rank, but the above conditions are satisfied, then the criterion is given by replacing all inverses in (3.6) by generalized inverses. Thus criterion (3.6) is applicable in linear models of less than full rank, as in analysis of variance and analysis of covariance.

3.1 Some Simplification

It is shown in the appendix that when M is chosen as in (2.5), criterion (3.6) can be written in the simpler form

$$\begin{aligned}
 (3.7) \quad & (y - X_1 \tilde{\beta}_1)' N (y - X_1 \tilde{\beta}_1) - \hat{\sigma}^2 (n - r) \\
 & + 2\hat{\sigma}^2 \text{trace} \left[(\tilde{X}'_1 \tilde{X}_1)^{-1} M_{11} \right] \\
 = & (y - X_1 \tilde{\beta}_1)' (I - Z(Z'Z)^{-1}Z') (y - X_1 \tilde{\beta}_1) - \hat{\sigma}^2 (n - r) \\
 & + 2\hat{\sigma}^2 \text{trace} \left[[X'_1 (I - Z_1(Z'_1 Z_1)^{-1} Z'_1) X_1]^{-1} \right. \\
 & \left. \cdot [X'_1 (I - Z(Z'Z)^{-1}Z') X_1] \right]
 \end{aligned}$$

where $N = (I_n - Z(Z'Z)^{-1}Z')$, and $\tilde{\beta}_1$ is the least-squares estimate of β_1 under the approximating model. Note that representation (3.6) is only a function of the least-squares estimates of β and γ under the operating model, so that least-squares estimation under the approximating model need not be performed to obtain the criterion. In contrast, to compute version (3.7) one needs the least-squares estimate for β_1 under the approximating model in question.

3.2 Three Special Cases

(i) $q_2 = 0$:

Here all the parameters β whose estimation is of interest are fitted in the approximating model, and the problem is to select a subset of the parameters γ . This situation was considered by Linhart and Zucchini (1982 and 1986, ch. 9), with $M = I_q$. When $q_2 = 0$, then X_2 and β_2 vanish so that $X_1 = X$, $S_2 = Z_2$ and $\tau_2 = \gamma_2$. After some algebra, criterion (3.6) simplifies to

$$\begin{aligned}
 (3.8) \quad & \hat{\gamma}'_2 Z'_2 \tilde{X} (\tilde{X}' \tilde{X})^{-1} M (\tilde{X}' \tilde{X})^{-1} \tilde{X}' Z_2 \hat{\gamma}_2 + 2\hat{\sigma}^2 \text{trace}[(\tilde{X}' \tilde{X})^{-1} M] \\
 & - \hat{\sigma}^2 \text{trace}[(\tilde{X}' \tilde{X} - \tilde{X}' Z_2 (Z'_2 Z_2)^{-1} Z'_2 \tilde{X})^{-1} M].
 \end{aligned}$$

If $M = X'X - X'Z(Z'Z)^{-1}Z'X = \tilde{X}'\tilde{X} - \tilde{X}'Z_2(Z'_2 Z_2)^{-1}Z'_2 \tilde{X}$ as in (2.5), criterion (3.8) simplifies further to

$$\hat{\gamma}'_2 Z'_2 \tilde{X} (\tilde{X}' \tilde{X})^{-1} M (\tilde{X}' \tilde{X})^{-1} \tilde{X}' Z_2 \hat{\gamma}_2 - \hat{\sigma}^2 q + 2\hat{\sigma}^2 \text{trace}[(\tilde{X}' \tilde{X})^{-1} M].$$

(ii) $q_1 = 0$ and $r_2 = 0$:

All the parameters β of interest are "estimated" to be zero, but the full vector γ is fitted. Here X_1, β_1, Z_2 and γ_2 vanish, so that $X_2 = X = S_2$, $\beta_2 = \beta = \tau_2$ and $Z_1 = Z = S_1$. Comparing the approximating model where the parameters of interest, β , are dropped ($\beta = 0$) with the operating model where they are fitted may be viewed as an alternative to testing the hypothesis $H_0 : \beta = 0$. The criterion for the operating model is

$$(3.9) \quad \text{trace}[\widehat{\text{cov}}(\tilde{\beta})M] = \hat{\sigma}^2 \cdot \text{trace}[(X'X - X'Z(Z'Z)^{-1}Z'X)^{-1}M]$$

and the criterion (3.6) for the approximating model simplifies to

$$(3.10) \quad \tilde{\beta}'M\tilde{\beta} - \hat{\sigma}^2 \cdot \text{trace}[(X'X - X'Z(Z'Z)^{-1}Z'X)^{-1}M].$$

One chooses the approximating model if (3.9) is smaller than (3.10). When M is chosen as in (2.5), then the approximating model is selected if

$$F = \tilde{\beta}'M\tilde{\beta}/q\hat{\sigma}^2 < 2$$

where F is the usual F-statistic associated with the hypothesis $H_0 : \beta = 0$. On the comparison of nested models using hypothesis tests see also Linhart and Zucchini (1986, pp. 117-118).

(iii) $r = 0$:

Here the estimation of the whole parameter vector is of interest. Thus Z and γ vanish. When M is chosen as in (2.4) or (2.5), criterion (3.6) simplifies to

$$(3.11) \quad \tilde{\beta}'_2(X'_2X_2 - X'_2X_1(X'_1X_1)^{-1}X'_1X_2)\tilde{\beta}_2 + \hat{\sigma}^2 \cdot (2p - k).$$

The quadratic form appearing in (3.11) is given by $RSS_p - RSS_k$, the difference between the residual sum of squares for the approximating and the operating models. Thus (3.11) is the criterion of Linhart and Zucchini (1986, p. 128) for linear operating and approximating models. Further, since $RSS_k = (n - k) \cdot \hat{\sigma}^2$, we have that (3.11) is given by $RSS_p + \hat{\sigma}^2(2p - n) = \hat{\sigma}^2 C_p$, and is therefore equivalent to Mallows' (1973) C_p criterion. Thus the criteria of Mallows (1973) and Linhart and Zucchini (1986) are special cases of (3.6). Note the equivalence of the Gauss-discrepancy, and the discrepancy based on the mean-squared error of prediction which leads to the C_p -criterion (see Linhart and Zucchini, 1986, p. 118).

3.3 Relationship to Cook's Distance

Another special case is given when $r_1 = 0$ and $q_2 = 0$, which is a special case of (i) above. Here all parameters whose estimation is of interest are fitted in the approximating model, and none of the others. Thus X_2, β_2, Z_1 and γ_1 vanish, so that $\tilde{X} = X, Z_2 = Z$ and $\gamma_2 = \gamma$ in (3.8) yields the criterion for this approximating model. One chooses the approximating model if (3.8) is smaller than (3.10), the criterion for the operating model, that is if

$$(3.12) \quad \hat{\gamma}' Z' X (X' X)^{-1} M (X' X)^{-1} X' Z \hat{\gamma} / q \hat{\sigma}^2 \\ < 2 \cdot \text{trace}[(X' X - X' Z (Z' Z)^{-1} Z' X)^{-1} M - (X' X)^{-1} M] / q.$$

Noting that $(X' X)^{-1} X' Z \hat{\gamma}$ is the difference between the least-squares estimates for β under the operating and the approximating model, it follows that the left hand side of (3.12) is $C_{(Z)} = C_{(Z)}(M, \hat{\sigma}^2)$, namely Cook's distance for the variable Z as defined by Schall and Dunne (1987). The statistic $C_{(Z)}(M, c)$ reduces to the statistic $D_i(M, c)$ of Cook and Weisberg (1982, p. 114) when Z is a dummy indicator variable for the i -th observation. Note that choices (2.4) and (2.5) of M have been discussed by Cook and Weisberg (1982, p. 124) for their statistic $D_i(M, c)$, and by Schall and Dunne (1987) for the general case. Two possible choices of c are the residual mean-square under the approximating model,

and the residual mean-square under the operating model. The latter is used above. When M is chosen as in (2.4), then the right hand side of (3.12) is given by $2.(HVF_{(Z)}(X)/q - 1) = 2.(trace[(X'X - X'Z(Z'Z)^{-1}Z'X)^{-1}(X'X)]/q - 1)$, where $HVF_{(Z)}(X)$ is the generalized variance inflation factor for X and Z based on Hotelling's trace statistic, as defined by Schall and Dunne (1988, eq. 3.7). When M is chosen as in (2.5), then the right hand side of (3.12) is given by $2.(1 - PVIF_{(Z)}(X)/q) = 2.(1 - trace[(X'X)^{-1}(X'X - X'Z(Z'Z)^{-1}Z'X)]/q)$, where $PVIF_{(Z)}(X)$ is the generalised variance inflation factor for X and Z based on Pillai's trace statistic, as defined by Schall and Dunne (1988, eq. 3.8). Note that for $q = 1$ both generalized variance inflation factors reduce to the well-known variance inflation factor of Marquardt (1970). For the two choices of M discussed above, condition (3.12) can be written as

$$(3.13) \quad \begin{aligned} C_{(Z)} &< 2.[HVF_{(Z)}(X)/q - 1] \\ C_{(Z)} &< 2.[1 - PVIF_{(Z)}(X)/q]. \end{aligned}$$

Thus the variables Z should be dropped from the operating model if Cook's distance for the variables is smaller than a simple function of the variance inflation factor associated with the variables. The implications of relationship (3.13) for the treatment of influential observations in linear models will be discussed elsewhere.

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APPENDIX: PROOF OF EQUATION (3.7)

Note that

$$\begin{aligned} \text{(A.1)} \quad & (y - X_1 \tilde{\beta}_1)' N (y - X_1 \tilde{\beta}_1) \\ &= (y - X \hat{\beta} + X \hat{\beta} - X_1 \tilde{\beta}_1)' N (y - X \hat{\beta} + X \hat{\beta} - X_1 \tilde{\beta}_1) \\ &= (n - k) \cdot \hat{\sigma}^2 + \tilde{b}' M \tilde{b} \end{aligned}$$

which allows one to rewrite the first term in (3.6). Further we have

$$\text{(A.2)} \quad B(S_2' S_2 - S_2' S_1 (S_1' S_1)^{-1} S_1' S_2)^{-1} B' = M^{-1} - \begin{bmatrix} (\tilde{X}_1' \tilde{X}_1)^{-1} & 0 \\ 0 & 0 \end{bmatrix}.$$

Using (A.1) and (A.2) in (3.6) leads to (3.7). To show (A.2), use a well-known matrix inversion formula (see Rao, 1973 p.35) to write

$$\begin{bmatrix} S'_1 S_1 & S'_1 S_2 \\ S'_2 S_1 & S'_2 S_2 \end{bmatrix}^{-1} = \begin{bmatrix} (S'_1 S_1)^{-1} + F E^{-1} F' & -F E^{-1} \\ -E^{-1} F' & E^{-1} \end{bmatrix}$$

where $E = S'_2 S_2 - S'_2 S_1 (S'_1 S_1)^{-1} S'_1 S_2$ and $F = (S'_1 S_1)^{-1} S'_1 S_2$. Now

$$\begin{aligned} (A.3) \quad M^{-1} &= (X'X - X'Z(Z'Z)^{-1}Z'X)^{-1} \\ &= [I_q : 0] \begin{pmatrix} X'X & X'Z \\ Z'X & Z'Z \end{pmatrix}^{-1} \begin{pmatrix} I_q \\ 0 \end{pmatrix} \\ &= \begin{bmatrix} I_{q_1} & 0 & 0 & 0 \\ 0 & 0 & I_{q_2} & 0 \end{bmatrix} \begin{bmatrix} F E^{-1} F' & -F E^{-1} \\ -E^{-1} F' & E^{-1} \end{bmatrix} \begin{bmatrix} I_{q_1} & 0 \\ 0 & 0 \\ 0 & I_{q_2} \\ 0 & 0 \end{bmatrix} \\ &\quad + \begin{bmatrix} [I_{q_1} : 0] (S'_1 S_1)^{-1} \begin{bmatrix} I_{q_1} \\ 0 \end{bmatrix} & 0 \\ 0 & 0 \end{bmatrix} \\ &= B(S'_2 S_2 - S'_2 S_1 (S'_1 S_1)^{-1} S'_1 S_2)^{-1} B' + \begin{bmatrix} (\tilde{X}'_1 \tilde{X}_1)^{-1} & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

To show the last equality in (A.3) note that

$$B = \begin{bmatrix} [I_{q_1} & : 0] F' \\ -I_{q_2} & : 0 \end{bmatrix}.$$

But (A.3) shows (A.2), which together with (A.1) proves the equivalence of (3.6) and (3.7), as pointed out above.